

REVENUE DECLINE INVESTIGATION

Why has Revenue declined by 43.1%?

Dataset: Contoso Retail Sales

This project investigates the decline across two areas: Product and Geography. A drill-down approach is used to identify the root causes and understand the key contributors to the decline. Additionally, Financial Impact and Recovery Analysis were conducted to assess the overall business performance.

The goal is to identify the contributing factors, understand the financial impact of the decline, and pinpoint areas of recovery, so the business can understand what led to the decline.

Note: The dataset does not contain data for January, February, March, November, and December. Therefore, the analysis focused on the available months from April to October.

Overview

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- 03 Financial Impact
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About the Project

TOOLS SQL | Power BI | MS-Excel

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Executive Overview

Introduction

Executive Overview

Root-cause Analysis

Financial Impact

Recovery Analysis

Recommendations

Increase

Decrease

Neutral

Select Category

All

Select Subcategory

All

Select Continent

All

Select Country

All

Clear all slicers

Revenue %YoY

-43.1% ▼

Revenue 2011 \$152.2M

Revenue 2012 \$86.6M

Revenue Change -\$65.6M

Gross Profit %YoY

-43.7% ▼

Gross Profit 2011 \$89.3M

Gross Profit 2012 \$50.3M

Gross Profit Change -\$39.0M

Quantity Sold %YoY

-41.9% ▼

Quantity Sold 2011 442,660

Quantity Sold 2012 257,330

Quantity Sold Change -185,330

Avg. Selling Price %YoY

-2.2% ▼

ASP 2011 \$343.8

ASP 2012 \$336.4

ASP Change -\$7.4

Revenue Trend Analysis

2011 2012

Revenue

Gross Profit

Quantity Sold

ASP

\$0.5M

\$0.0M

Apr

May

Jun

Jul

Aug

Sep

Oct



Apr, May, June, July and October are selected for further investigation based on their contribution to the Revenue and Quantity Sold decline.

Revenue Recovery

Apr

-46.3%

May

-47.6%

Jun

-49.2%

Jul

-43.5%

Aug

5.6%

Sep

3.4%

Oct

-40.2%

-40%

-20%

0%

Monthly Revenue Contribution %YoY

Month

%Revenue

%Gross Profit

%Quantity Sold

%Contribution by Month

Apr

-46.3%

-47.1%

-42.7%

-19.9%

May

-47.6%

-48.1%

-47.0%

-27.6%

Jun

-49.2%

-49.7%

-48.0%

-27.5%

Jul

-43.5%

-44.5%

-42.8%

-4.3%

Aug

5.6%

5.4%

15.8%

0.9%

Sep

3.4%

1.9%

19.2%

1.1%

Oct

-40.2%

-40.4%

-41.9%

-22.8%

Total

-43.1%

-43.7%

-41.9%

-100.0%

Root-Cause Analysis

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Increase

Decrease

Neutral

Quantity Sold Change

-185K

Bottom 1 Class

103%

% of Decline

Bottom 5 Categories

98.8%

% of Decline

Bottom 9 Subcategories

86.7%

% of Decline

Bottom 4 Manufacturers

79%

% of Decline

Product

Geography



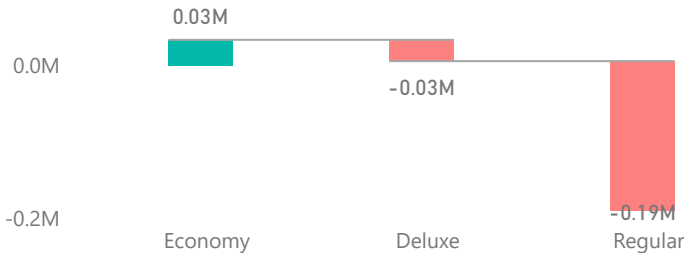
Each chart below drills into the previous one: Class to Category, Category to Subcategory, and Subcategory to Manufacturer, using the bottom performing segments at each level.

Quantity Change by Class

1

(within Apr, May, Jun, Jul & Oct)

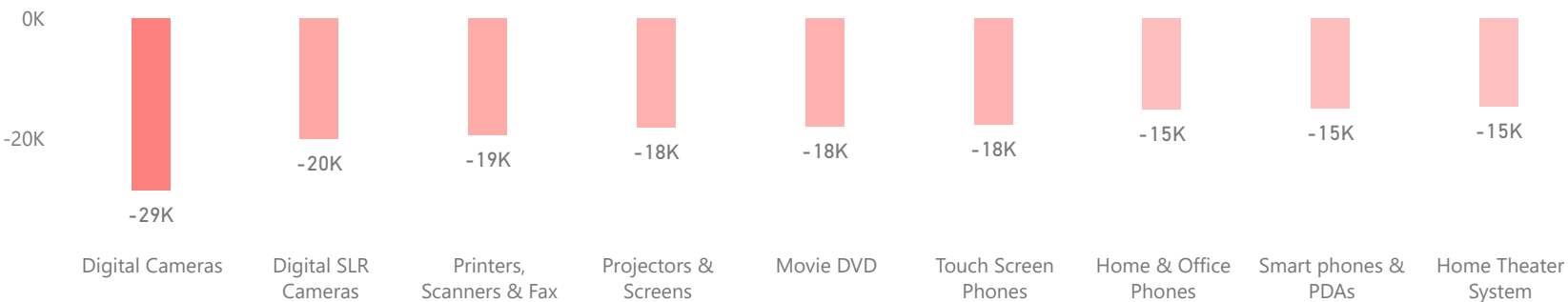
● Increase ● Decrease



Quantity Change of Bottom 9 Subcategories

3

(within previous 5 Categories)

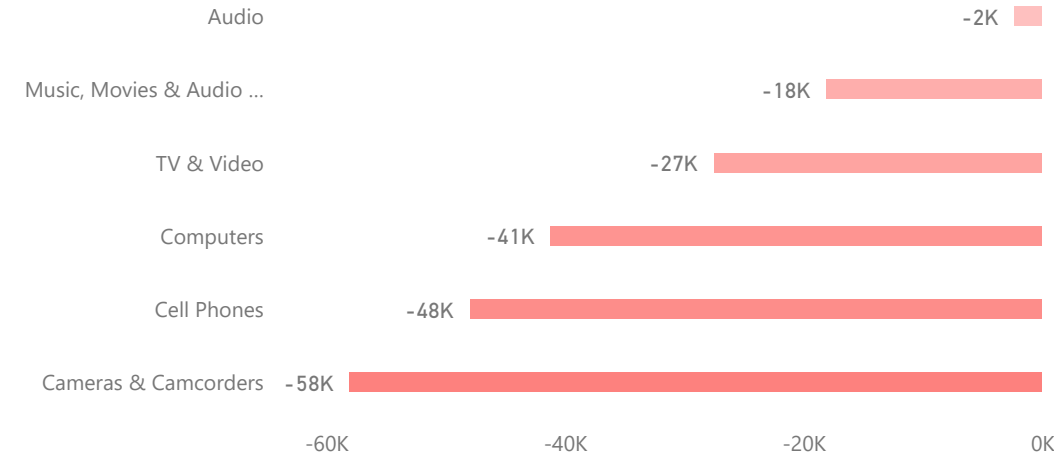


Quantity Change by Category

Details

Visual

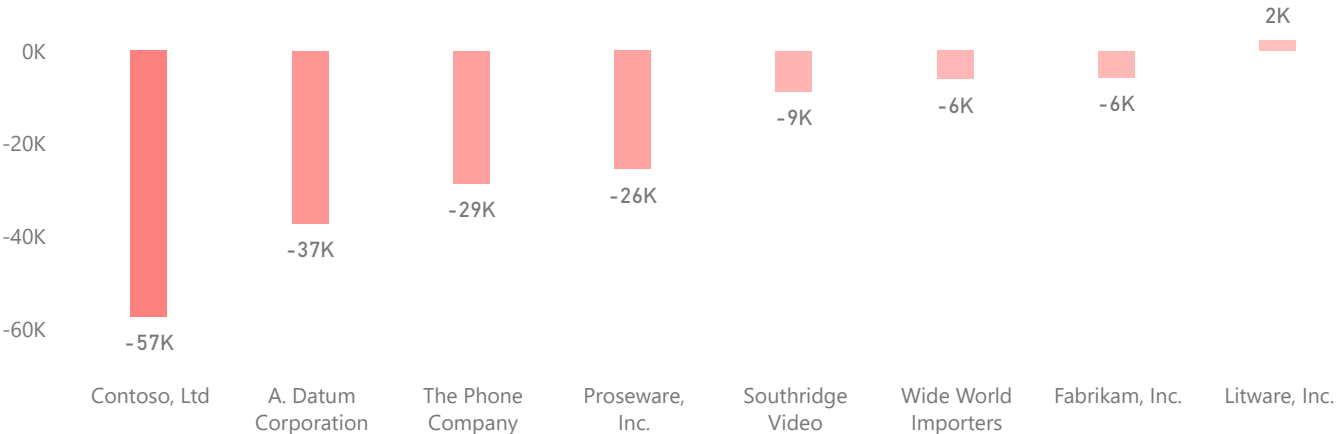
2



Quantity Change by Manufacturer

4

(within previous 9 Subcategories)



Financial Impact

Revenue %YoY-43.1% ▼

Revenue 2011	\$152.2M
Revenue 2012	\$86.6M
Revenue Change	-\$65.6M

Gross Profit %YoY-43.7% ▼

Profit 2011	\$152.2M
Profit 2012	\$86.6M
Profit Change	-\$65.6M

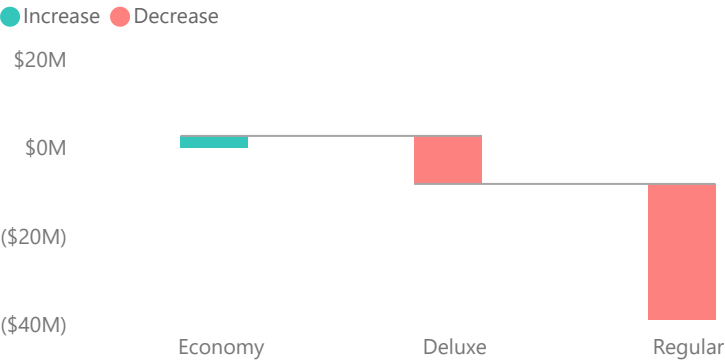
Cost %YoY-42.4% ▼

Cost 2011	\$62.9M
Cost 2012	\$36.3M
Cost Change	-\$26.7M

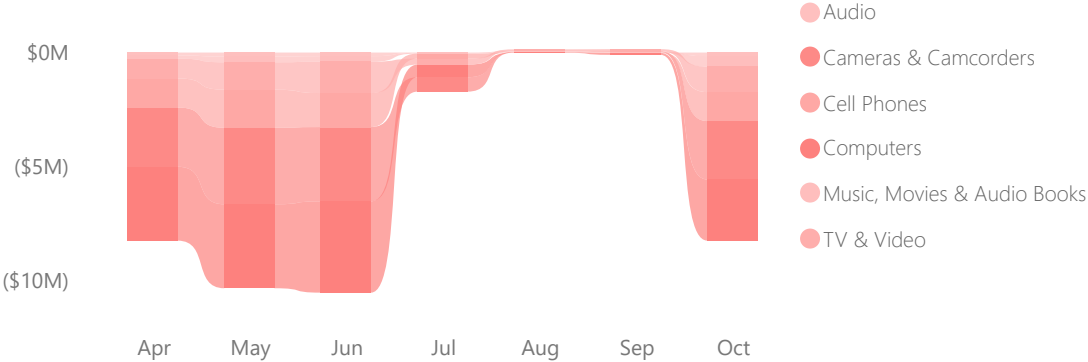
Profit Margin %YoY-42.6% ▼

Profit Margin 2011	\$2.5M
Profit Margin 2012	\$1.4M
Profit Margin Change	-\$1.1M

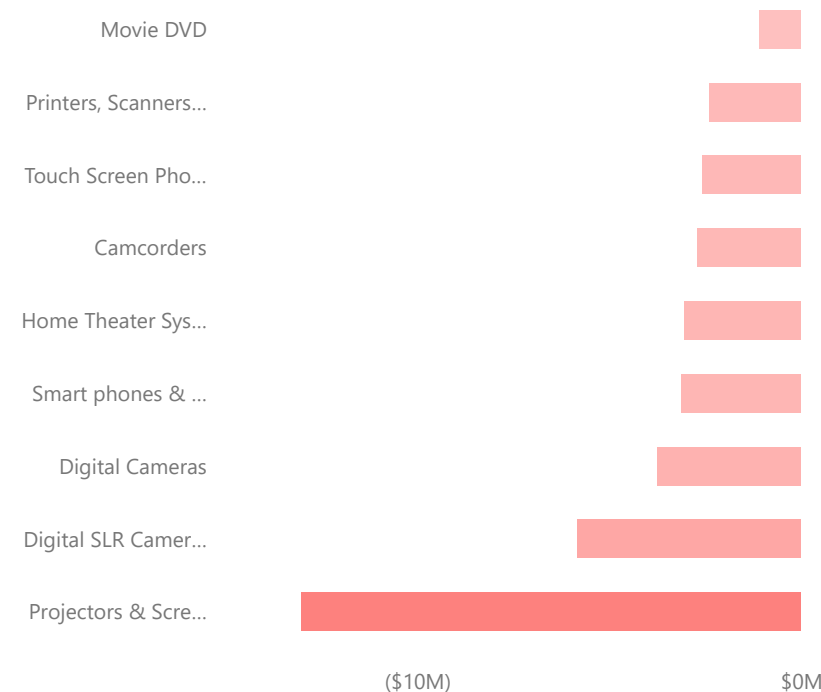
Profit Change by Product Class



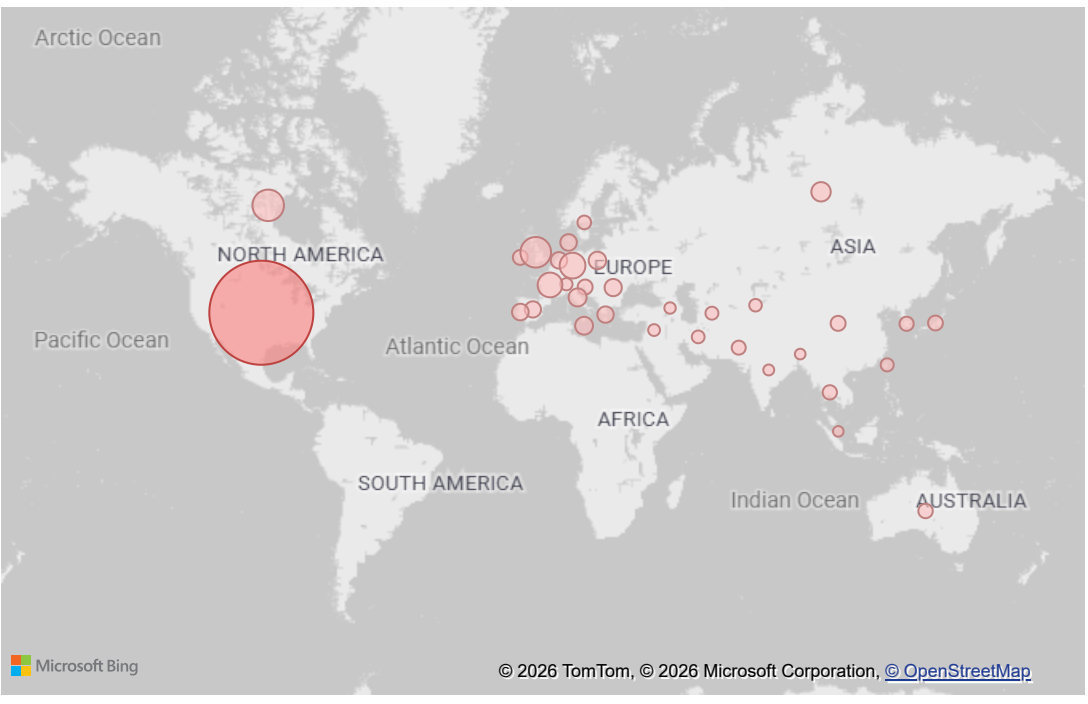
Profit Change Trend by Product Category



Profit Change by Bottom 9 Product Subcategories



Profit Change by Country



Recovery Analysis

Product

Geography & Store



Each chart below drills into the previous one: Class to Category, Category to Subcategory and Subcategory to Manufacturer, using the top performing segment at each level.

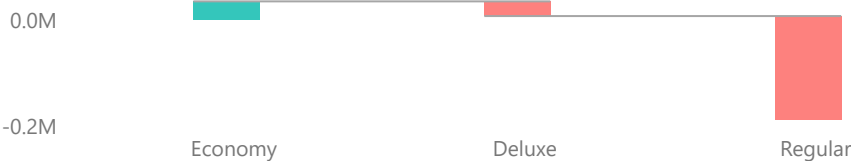
Revenue %YoY-79.9%▼

Revenue 2011	\$148.1M
Revenue 2012	\$82.3M
Revenue Change	-\$65.8M

Quantity Change by Product Class

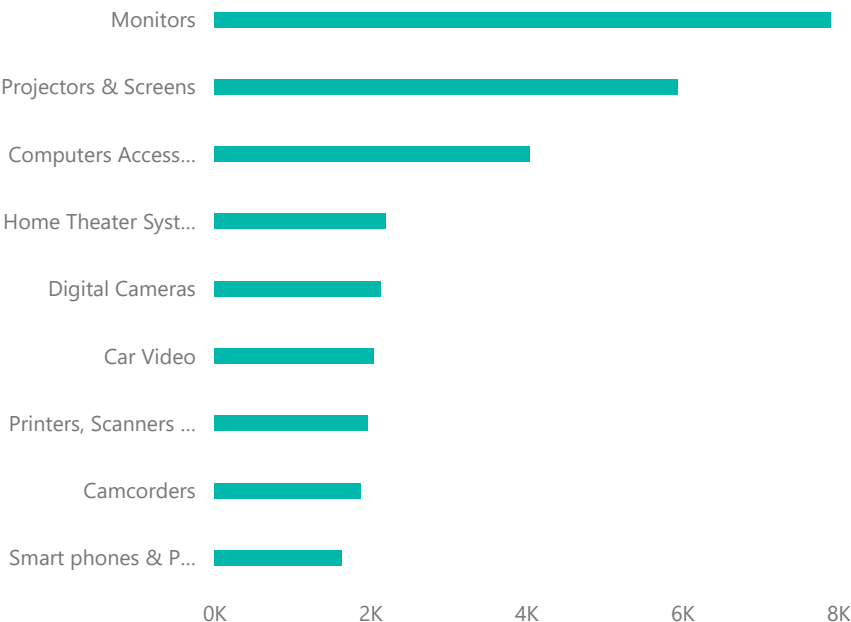
(Apr, May, Jun, Jul & Oct)

● Increase ● Decrease



Quantity Change by Top 9 Subcategories

(within previous 4 Categories)

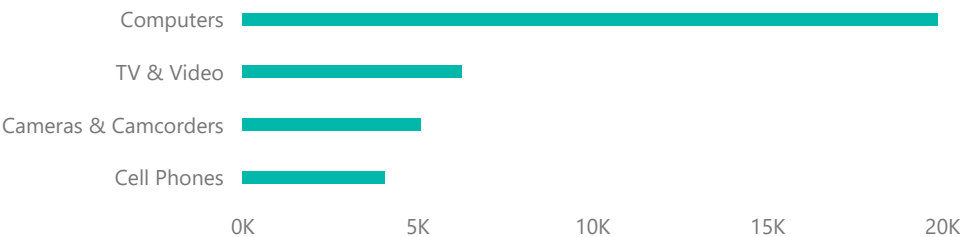


Quantity Sold %YoY-43.4%▼

Quantity Sold 2011	431,750
Quantity Sold 2012	244,320
Quantity Sold Change	-187,430

Quantity Sold Recovery by Category

(within Economy)

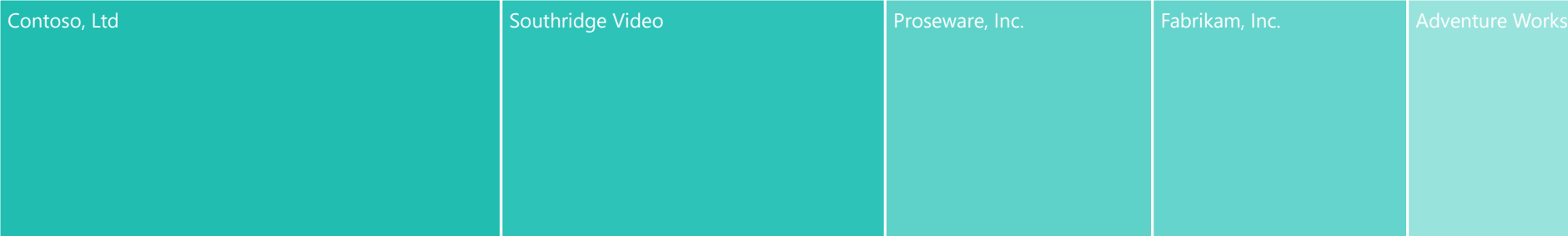


Avg. Selling Price %YoY-4.4%▼

ASP 2011	\$334.6
ASP 2012	\$319.9
ASP Change	(\$14.6)

Quantity Change by Top 5 Manufacturers

(within previous 9 Subcategories)



 Limited available data. Some recommendations require further investigation & validation before action.

Revenue %YoY -43.1% ▼

Revenue 2011	\$152.2M
Revenue 2012	\$86.6M
Revenue Change	-\$65.6M

Gross Profit %YoY -43.7% ▼

Profit 2011	\$89.3M
Profit 2012	\$50.3M
Profit Change	-\$39.0M

Quantity Sold %YoY -41.9% ▼

Quantity 2011	442,660
Quantity 2012	257,330
Quantity Change	-185,330

Root-cause & Financial Impact



CAUSE

MONTHLY TREND

102% of quantity decline occurred in Apr-Jul & Oct

PRODUCT CLASS

103% came from Regular Product Class

PRODUCT CATEGORY

98.8% came from 5 categories:
(Cameras & Camcorders, Cell Phones, Computers, TV & Video, Music, Movies & Audio Books)

PRODUCT SUBCATEGORY

86.7% came from 9 subcategories:
(Digital Cameras, Digital SLR Cameras, Printers, Scanners & Fax, Projectors & Screens, Movie DVD, Touch Screen Phones, Home & Office Phones, Smart Phones & PDAs, Home Theater System)

MANUFACTURER

79% came from 4 manufacturers:
(Contoso, A Datum Corporation, The Phone Company, Proseware)

GEOGRAPHY

85% came from North America, with 93.5% from the US

Recovery Analysis



IMPACT

REVENUE

43% decline occurred in 1 year

QUANTITY SOLD

41.9% decline occurred because of the revenue (primary driver)

GROSS PROFIT

43.7% decline occurred during that 1 year

COST

42.4% cost reduced during 1 year (not the primary driver)

Further Investigation



ACTION

UNDERSTAND SEASONAL TREND

Identify seasonal demand patterns and plan inventory accordingly.

DEMAND DRIVERS & PRICING STRATEGY

Investigate demand drivers and optimize pricing

FOCUS ON PRODUCT MIX

Adjust product mix based on market conditions, customer preferences, and competitive behavior.

HIGH MARGIN PRODUCTS

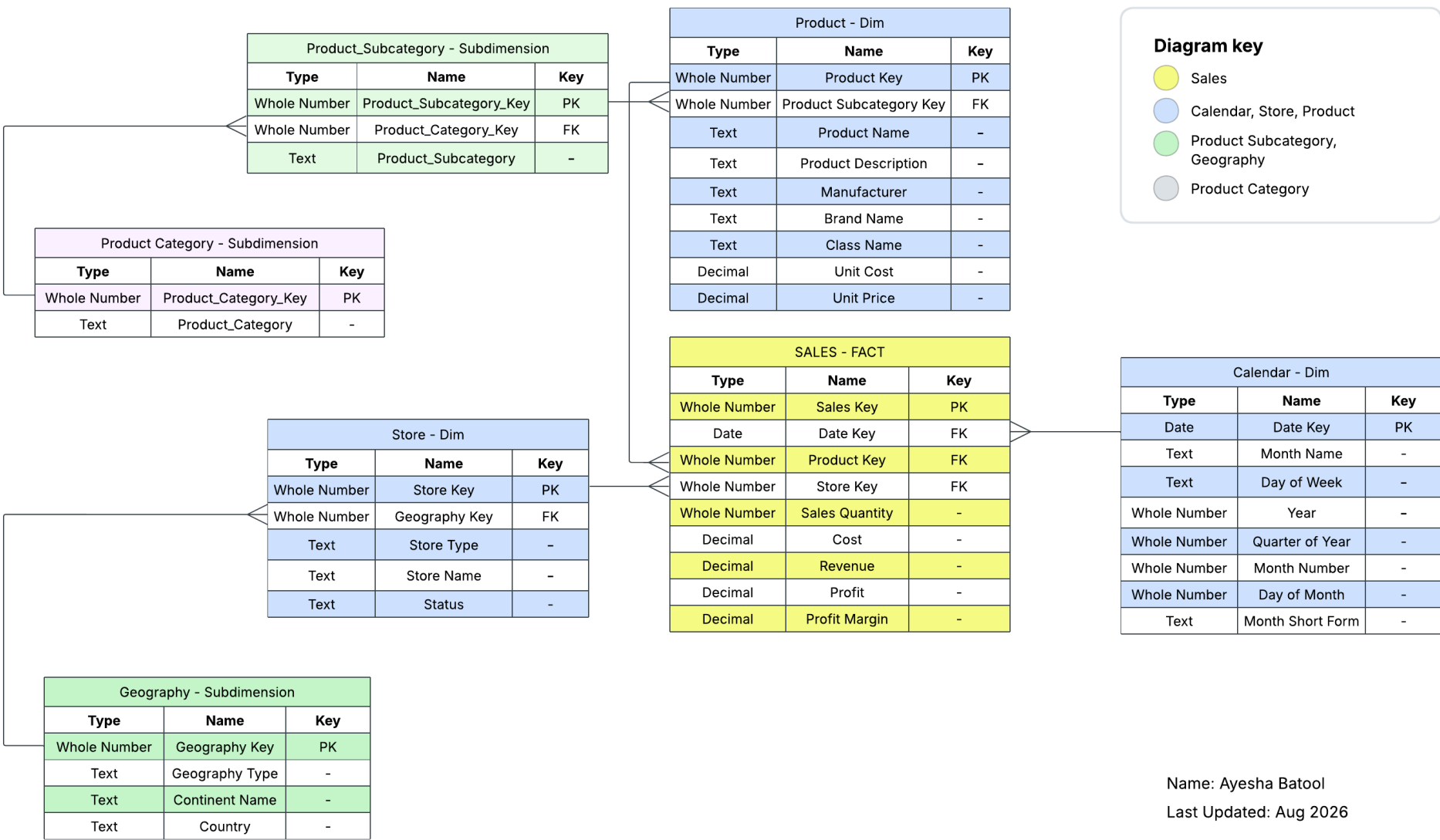
Identify products with high demand and high profit margin to improve business health

CUSTOMER ACQUISITION, RETENTION AND MARKETING STRATEGY

Measure customer acquisition & retention rates to identify declining customer segments & refine marketing strategies.

ENTITY RELATIONSHIP DIAGRAM (ERD)

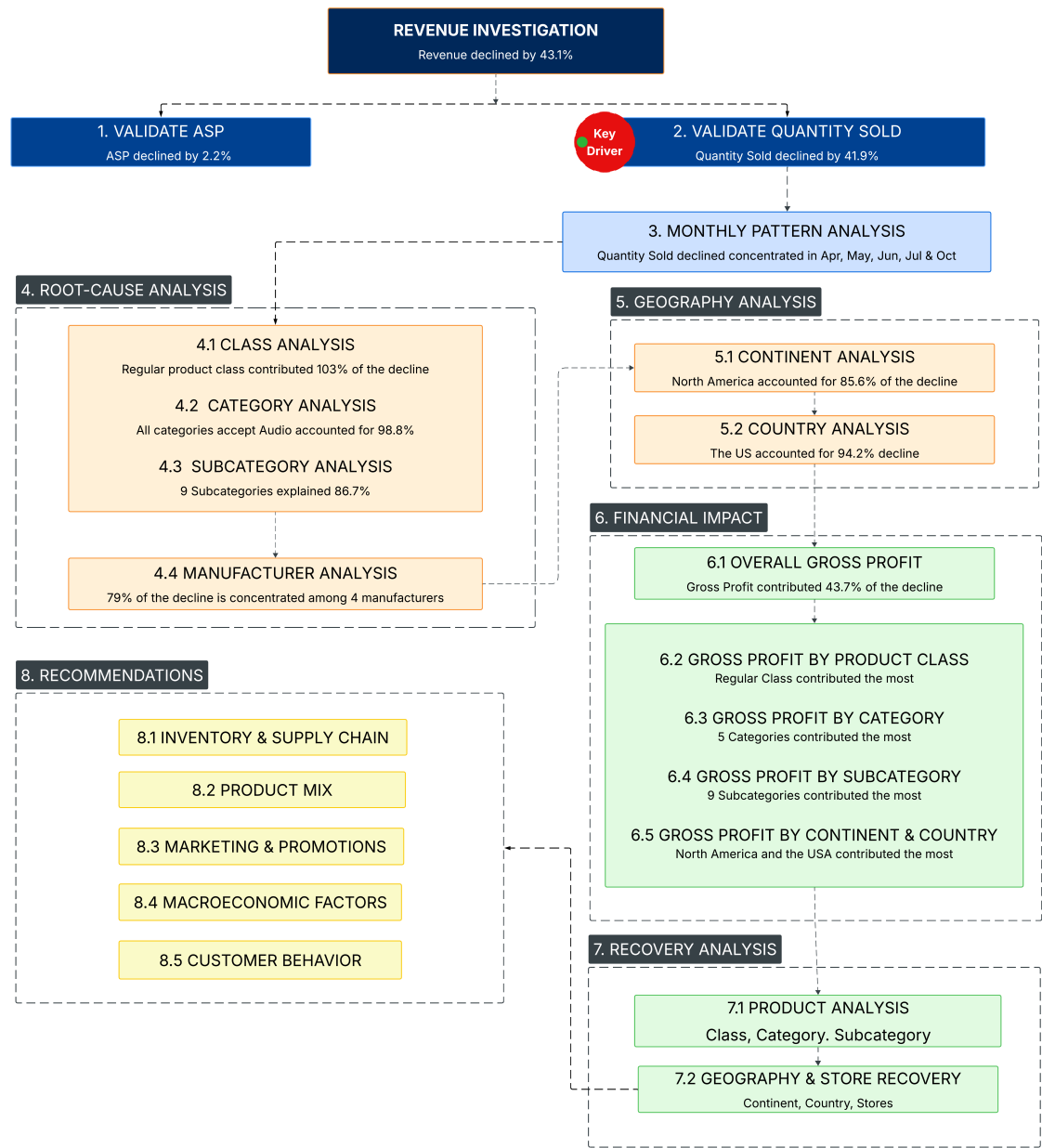
Revenue Decline Analysis



DATA DICTIONARY

Table Name ▼	Column Name	Data Type	Description	Key
Stores	Geography_Key	Whole Number	Links each store to its corresponding geographic location.	FK
Stores	Status	Text	Operational status of the store (e.g., On or Off).	-
Stores	Store_Key	Whole Number	Unique identifier for each store.	PK
Stores	Store_Name	Text	Name of the store.	-
Stores	Store_Type	Text	Type or format of the store	-
Sales	Date_Key	Date	Links each sales transaction to the Calendar table	FK
Sales	Product_Key	Whole Number	Links each sales transaction to the Product table	FK
Sales	Profit	Decimal Number	Total Profit earned from the sales transaction	-
Sales	Profit_Margin	Decimal Number (%)	Profit expressed as a percentage of revenue for a sales transaction	-
Sales	Revenue	Decimal Number	Total Revenue generated from the sales transaction	-
Sales	Sales_Key	Whole Number	Unique identifier for each sales transaction	PK
Sales	Sales_Quantity	Whole Number	Number of units sold in each sales transaction	-
Sales	Store_Key	Whole Number	Links each sales transaction to the Store table	FK
Product_Subcategory	Product_Category_Key	Whole Number	Name of the product subcategory	FK
Product_Subcategory	Product_Subcategory	Text	Name of the product subcategory	-
Product_Subcategory	Product_Subcategory_Key	Whole Number	Unique identifier for each product subcategory.	PK
Product_Category	Product_Category	Text	Name of the product category	-
Product_Category	Product_Category_Key	Whole Number	Unique identifier for each product category.	PK
Product	Brand_Name	Text	Brand under which the product is sold	-
Product	Class_Name	Text	Product classification	-
Product	Manufacturer	Text	Name of the product manufacturer	-
Product	Product_Description	Text	Detailed description of the product	-
Product	Product_Key	Whole Number	Unique identified for each product	PK
Product	Product_Name	Text	Name of the Product	-
Product	Product_Subcategory_Key	Whole Number	Links each product to its corresponding product subcategory	FK
Product	Unit_Cost	Decimal Number	Cost incurred per unit of the product	-
Product	Unit_Price	Decimal Number	Selling Price per unit of the product	-
Geography	Continent_Name	Text	Name of the continent where the location is situated.	-
Geography	Country	Text	Name of the country associated with the geographic location.	-

ANALYTICAL WORKFLOW



PROJECT WORKFLOW

